



Alotam
परिभाषा नियन्त्रक



FAR-WESTERN UNIVERSITY
End-Term Examination-2075
Parallel Computing (CSIT.421)

Faculty: Science and Technology (CSIT)
Level: Undergraduate
Semester: Eighth

Full Marks: 80

Time: 2hrs. 40 minutes

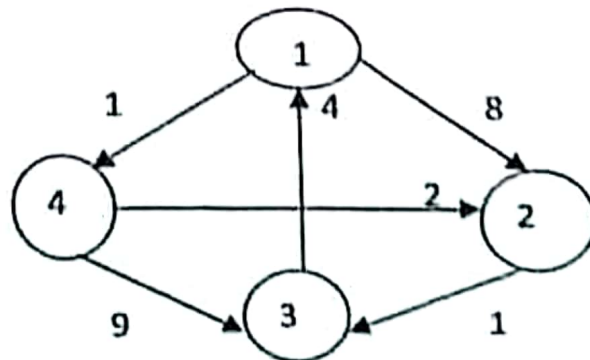
Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Attempt any six questions.

Group - B

6×8 = 48

1. How does OpenMP provide a shared-memory programming environment? What is the memory consistency model supported by OpenMP?
2. What is Routing mechanism? Differentiate between deterministic routing and adaptive routing.
3. What is pipelining? Describe the speed up gain due to pipelining.
4. What do you mean by parallel reduction operation? Differentiate between single reduction and multiple reduction operations on a single set of data.
5. What is hypercube connection? Explain forward and reverse phase of prefix computation on hypercube.
6. Define Amdahl's law? What do you concluded from Amdahl's law?
7. Write short notes on (any two)
 - a) MIMD
 - b) Task Allocation
 - c) Monte Carlo method
8. Write Floyd-Warshall algorithm. Use this algorithm to find the shortest path between all pairs of vertices of figure given below:



P.T.O.

FAR WESTERN UNIVERSITY
Semester End Examination-2080
Parallel Computing (CSIT.421)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: Eighth

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Full Marks: 100
Time: 3hrs.

Group – 'A'

8×3 = 24

Attempt all questions (Very short answer questions)

1. How will you define speed up if a number of processors is added to perform fraction of work in parallel?
2. How does OpenMP provide a shared-memory programming environment?
3. What are the classifications of routing mechanism?
4. Differentiate between data and task parallelism.
5. Define the term agglomeration.
6. Explain Monte Carlo method.
7. What is NUMA memory architecture?
8. Explain about loop Jamming.

Group – 'B'

5×8=40

Attempt any five questions. (Short answer questions)

9. Explain the concept of speed up by applying Amdahl's law. What is speed up if no part of the code can be parallelized? What is the speed up if 50% of the code can be parallelized?
10. Why do we use MPI? Discuss basic concept of the MPI.
11. Write Floyd-Warshall algorithm. Trace it with suitable example.
12. Explain Hyper quicksort algorithm with example.
13. Explain in detail about Flynn's classification.
14. How can you solve linear system? Use Jacobi iteration method to solve the following equations:
$$\begin{aligned} 6x_1 - 2x_2 + x_3 &= 11 \\ x_1 + 2x_2 - 5x_3 &= -1 \\ -2x_1 + 7x_2 + 2x_3 &= 5 \end{aligned}$$

Group – 'C'

Attempt any three questions. (Long answer questions)

3×12 = 36

15. Explain Karp-Flatt metric. Consider the following speed ups obtained by a certain parallel application:

No. of processors	2 CPUs	3 CPUs	4 CPUs	5 CPUs	6 CPUs	7 CPUs	8 CPUs
ϕ	1.17	2.33	3.23	3.53	4.26	4.56	4.77

Find the value of 'e'?

16. Explain Canon's algorithm. Use Eratosthenes algorithm to find all the prime number between 1 to 40.
17. Explain the architecture of distributed and shared memory computer. What are the major differences between distributed and shared memory computers?
18. Define clustering. What are the differences between parallel computing, distributed computing and cloud computing?

FAR WESTERN UNIVERSITY
Semester End Examination-2079
Parallel Computing (CSIT.421)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: Eighth

Full Marks: 100

Time: 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group – A

Attempt all questions (Very Short Questions)

5 hr

8×3=24

10+6+1

1. Define pipelining. ✓
2. What is NUMA memory architecture? ✓
3. What is the memory consistency model supported by OpenMP?
4. Explain Isoefficiency matrix. .
5. Write down the application of Monte Carlo method. ✓
6. Explain Cannon's algorithm. ✓
7. What are the steps of Hyper Quicksort? (2)
8. Differentiate between deterministic routing and adaptive routing.

Group – B

Attempt any five questions. (Short Questions)

5×8=40

9. Define Amdahl's law? Write down the limitation of Amdahl's law.
10. Explain the steps of Foster's design methodology.
11. Define message passing model. What are the advantages and disadvantages of messaging passing model?
12. Differentiate between data and task parallelism. 130
13. Explain any one example of combinatorial search in detail.
14. Explain Karp-Flatt metric with suitable example.

Group – C

Attempt any three questions. (Long Questions)

3×12=36

15. What is parallel computing? Explain different paradigms of parallel computing. Also discuss merits and demerits of parallel computation.
16. Explain the architecture of distributed and shared memory computer. What are the major differences between distributed and shared memory computers?
17. Explain Flynn's classification of computer architectures with its block diagram. (25) see memory
18. Explain Floyd-Warshall algorithm with suitable example.

FAR-WESTERN UNIVERSITY
End-Term Examination-2074
Parallel Computing (CSIT.421)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: Eighth

Full Marks: 80

Time: 2hrs. 40 minutes

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Group - B

Attempt any six questions.

6×8 = 48

1. What is parallel computing? Who is using parallel computing?
2. Explain Gustafson Barsis's law with suitable example and compare it with Amdahl's law.
3. Why message passing model are used? What is included and not included in MPI standard?
4. What is shared memory model? Explain its uses.
5. What is data parallelism? Explain machine model and CU/PU overlap.
6. How can you solve linear system? Use Jacobi iteration method to solve the following equations:
$$\begin{aligned} 6x_1 - 2x_2 + x_3 &= 11 \\ x_1 + 2x_2 - 5x_3 &= -1 \\ -2x_1 + 7x_2 + 2x_3 &= 5 \end{aligned}$$
7. Explain divide and conquer technique. Write down the features of hypercube.
8. Write short notes on: (any two)
 - a. Isoefficiency Metric
 - b. Monte Carlo methods
 - c. Clustering

Group - C

Attempt any two questions.

2×16 = 32

9. Explain Canon's algorithm. Use Eratosthenes algorithm to find all the prime number between 1 to 40.
10. What do you mean by combinatorial search? Explain any one example in detail.
11. Explain Karp-Flatt metric. Consider the following speed ups obtained by a certain parallel application:

No. of processors	2	3	4	5	6	7	8
	CPUs	CPUs	CPUs	CPUs	CPUs	CPUs	CPUs
ρ	1.87	2.63	3.23	3.73	4.14	4.46	4.71

Find the value of 'e'?

FAR WESTERN UNIVERSITY
Final Examination-2077
Parallel Computing (CSIT.421)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: Eighth

Full Marks: 80

Time: 2hrs. 40 minutes

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group – B

Attempt any six questions.

6×8 = 48

1. What are the criteria that are used to evaluate the cost and performance of static interconnection networks?
2. Explain the features of parallel object oriented programming.
3. Explain Amdahl's law with suitable example. Why actual speed ups are always less in Amdahl's law?
4. Explain distributed shared memory system. Write its advantages.
5. What are parallel reduction algorithms? How they are used in PRAM architecture.
6. Define message passing. Use Eratosthenes algorithm to find all the prime number between 1 to 20.
7. Discuss different parallel approaches including data parallelism and control parallelism.
8. Write short notes on: **(any two)**
 - a. SIMD
 - b. Pipelining
 - c. Canon's algorithm

Group – C

Attempt any two questions.

2×16 = 32

9. Explain Karp-Flatt metric. Consider the following speed ups obtained by a certain parallel application:

No. of processors	2 CPUs	3 CPUs	4 CPUs	5 CPUs	6 CPUs	7 CPUs	8 CPUs
Ψ	1.73	2.81	3.11	3.57	4.00	4.38	4.82

Find the value of 'e'?

FAR-WESTERN UNIVERSITY
End-Term Examination-2076
Parallel Computing (CSIT.421)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: Eighth

Full Marks: 80

Time: 2hrs. 40 minutes

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group - B

Attempt any six questions.

6×8 = 48

1. What is NUMA memory architecture? Differentiate between one-to-all broadcast and all-to-one reduction.
2. Explain Gustafson Barsis's law and compare it with Amdahl's law.
3. What is PRAM Model? What are the subclasses of PRAM?
4. Differentiate between parallel and serial computing. Explain Foster's design methodology.
5. Explain Sieve of Eratosthenes algorithm. Use this algorithm to find all the prime numbers between 1 to 20.
6. What is parallel reduction operation? Explain its types.
7. Write Floyd-Warshall algorithm. Trace it with suitable.
8. What is data parallelism? Explain machine model and CU/PU overlap.

Group - C

Attempt any two questions.

2×16 = 32

9. What is pipelining? Describe the speed up gain due to pipelining.
10. Explain hypercube algorithm. Write down the features of hypercube. Explain forward and reverse phase of prefix computation on hypercube.
11. Explain Karp-Flatt metric. Consider the following speed ups obtained by a certain parallel application:

No. of processors	2 CPUs	3 CPUs	4 CPUs	5 CPUs	6 CPUs	7 CPUs	8 CPUs
ϕ	1.82	2.50	3.08	3.57	4.00	4.38	4.72

Find the value of 'e'?

FAR WESTERN UNIVERSITY
Semester End Examination-2079
E-Business and E-governance (CSIT.425.2)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: Eighth

Full Marks: 100

Time: 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group – A

Attempt all questions (Very Short Questions)

8×3=24

1. Differentiate between E-government and E-governance.
2. Why is E-governance important? shw
3. What is government to citizen (G2C)?
4. Define G2G interaction with example.
5. What are the main challenges and courses that control E-governance in Nepal? Lepm
6. Which country has an efficient E-governance system and how?
7. What is the performance NIC?
8. Define digital divide

Group – B

Attempt any five questions. (Short Questions)

5×8=40

9. Write about E-governance in health and education. -
10. Describe evolution of E-governance and its scope. -
11. Discuss present global trends of growth in E-governance.
12. Explain critical flow model of E-governance with suitable example. -
13. What changes are brought in the citizen- governance relationship through digital governance. .
14. Discuss E- Readiness briefly. -

Group – C

Attempt any three questions. (Long Questions)

3×12=36

15. Describe maturity model in briefly.
16. Explain the evolutionary stages in E- governance.
17. Describe data ware house with its advantages and disadvantages.
18. Describe strategies and policy of ICT act 2072.

FAR-WESTERN UNIVERSITY
End-Term Examination-2076
E-Business and E-Governance (CSIT.425.2)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: Eighth

Full Marks: 80

Time: 2hrs. 40 minutes

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group - B

Attempt any six questions.

6×8 = 48

1. Describe evaluation of e-governance in Nepal. Write e-government master plan.
2. Define data mining. Write advantages and disadvantages of data mining.
3. Explain GIDC and data centre.
4. Describe human and technological infrastructural preparedness for e-readiness.
5. What is e-governance? Why e-governance is necessary? Describe digital divide.
6. Explain present global trends in e-governance.
7. What is maturity model in e-governance? Describe in briefly.
8. What is cyber crime? Explain cyber law.

Group - C

Attempt any two questions.

2×16 = 32

9. What is data warehouse? Write application, advantages and disadvantages of data warehouse.
10. What is ICT Act? Mention objective and action plan of ICT Act.
11. Explain models of digital governance. Describe broadcasting dissemination, critical flow and comparative analysis model.

FAR-WESTERN UNIVERSITY
End-Term Examination-2074

E-Business and E-Governance (CSIT.425.2)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: Eighth

Full Marks: 80

Time: 2hrs. 40 minutes

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Group - B

Attempt any six questions.

6×8 = 48

1. Define e-governance. Why e-governance is necessary? Write advantages of e-governance.
2. What are the problems encountered in achieving e-governance in the context of Nepal?
3. Discuss the legal infrastructure preparedness of e-governance.
4. Write the application areas of data warehouse and data mining.
5. Describe the action plan of ICT policy 2072.
6. Explain the dissemination of information technology.
7. What is cyber law? Describe the needs of cyber law.
8. Explain maturity model.

Group - C

Attempt any two questions.

2×16 = 32

9. Describe data warehouse and data mining. Write advantages of data warehouse and data mining.
10. Describe the scope, content and present global trends of growth in e-governance.
11. What is model of digital governance? Explain any three model of digital governance.

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End-Term Examination-2075
E-Business and E-Governance (CSIT.425.2)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: Eighth

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Full Marks: 80

Time: 2hrs. 40 minutes

Group - B

6×8 = 48

Attempt any six questions.

- ① What is E-governance? Explain the advantages of e-governance.
- ② What are the key challenges of implementing E-governance in developing countries?
- ③ Describe the latest IT policy of Nepal in brief?
- ④ Explain the various levels of E-governance maturity modal?
- ⑤ Define data warehouse and data mining. Write down their advantages.
- ⑥ What is cyber law? What are the various areas that cyber law covers?
- 7) Define OLAP. What is the importance of OLAP in E-governance?
- 8) What is E-readiness? Discuss the legal infrastructure preparedness of E-governance.

Group - C

2×16 = 32

Attempt any two questions.

- 9) Explain six generic models of digital governance in developing countries with their applications.
- ⑩ Explain the evolutionary stages of e-governance that are perceived in any governmental organization.
- ⑪ If you are elected as a information technology minister, what are your master plans for implementing e- governance in Nepal. Explain in your own view.

Optimized code:

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printf("Pankaj\n");
printf("Pankaj\n");
printf("Pankaj\n");
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FAR WESTERN UNIVERSITY
Semester End Examination-2080

E-Business and E-Governance (CSIT.425.2)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: Eighth

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Group – 'A'

Attempt all questions (Very short answer questions)

8×3 = 24

1. What social impact does e-governance has on society?
2. List the advantages of E-Governance.
3. What is encryption? Write its types.
4. What is the use of digital signatures?
5. Which is the utilization of electronic technology in the administration?
6. What is the performance of NIC?
7. How can e-governance security difficulties can be withdrawn?
8. What is impact of e-Learning sites on students?

Group – 'B'

Attempt any five questions. (Short answer questions)

5×8=40

9. Explain the relation between ICT and governance.
10. Describe e-government maturity model.
11. Write the challenges of e-Governance with specific focus to Nepal.
12. Discuss the scope and content of e-governance.
13. Define cyber law. Why it is important for Nepal? Explain.
14. What is institutional infrastructural preparedness for e-readiness? Describe.

Group – 'C'

Attempt any three questions. (Long answer questions)

3×12 = 36

15. Why do we need e-governance? How can we use-governance to provide efficient services to citizens? Explain using suitable examples.
16. What is e-governance model? Explain critical flow model and comparative analysis model in details.
17. Define the terms data warehousing and data mining. Discuss key areas that demand the use of data warehousing and data mining.
18. Give the reason, why IT infrastructure is needed to implement e-governance. Explain network and computing infrastructure needed to implement e-governance.

FAR-WESTERN UNIVERSITY
End-Term Examination-2076
Cloud Computing (CSIT.424.4)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: Eighth

Full Marks: 80

Time: 2hrs. 40 minutes

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group - B

Attempt any six questions.

6×8 = 48

- 1) What do you mean by cloud? Describe about the basic characteristics of cloud.
- 2) Explain platform as a service (PaaS) and its characteristics.
- 3) Define virtualization. What is the role of virtualization in cloud computing.
- 4) Why Cloud Migration is needed? Explain model of migration into cloud.
- 5) Explain the service oriented architecture (SOA) and its characteristics.
- 6) Explain the service provided by Amazon EC2 services from user perspective.
- 7) Explain the cloud security challenges.
- 8) Write Short Notes on
 - A) Grid computing
 - B) Risk Assessment in cloud

Group - C

Attempt any two questions.

2×16 = 32

- 9) What type of development models can be adopted in cloud computing. Describe each of them with suitable example.
- 10) What do you mean by an intrusion in a cloud network? How intrusions in a cloud network are detected.
- 11) Describe about Jericho Cloud Cube Model. What are the advantages of Communication -as-a service (CaaS) Model?

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FAR WESTERN UNIVERSITY
Semester End Examination-2079
Cloud Computing (CSIT 424.4)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: Eighth

Full Marks: 100

Time: 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group – A

Attempt all questions (Very Short Questions)

Shu

8×3=24

1. Write short note on IaaS model ✓
2. Write in short on Public Cloud Computing Deployment Models. ✓
3. What are different advantages of CaaS? ✓
4. What are key characteristics of SaaS software? ✓
5. Explain in short single-purpose architectures to multi-purpose architectures. ✓
6. What are different approaches used for host security. ✓
7. What are different factors involved in disaster recovery planning. ✓
8. Explain in short Distributed Computing in Grid and Cloud. ✓

Group – B

Attempt any five questions. (Short Questions)

5×8=40

9. Explain the concept of implementing network intrusion detection in the cloud. ✓
10. What are the different approaches for host security in cloud computing? ✓
11. What are the benefits of data center virtualization? ✓
12. What are the principal characteristics of SOA? ✓
13. Explain CaaS with its advantages in short. ✓
14. Explain Amazon EC2 service with its characteristics. ✓

Group – C

Attempt any three questions. (Long Questions)

3×12=36

15. Explain the term cloud computing with its characteristics and explain difference between grid computing and cloud computing. ✓
16. What is data center? Explain in detail with its building blocks. ✓
17. Explain cloud security challenges and explain why we need cloud security requirements and also explain what are security issues which one should discuss with a cloud-computing vendor. ✓
18. What are different techniques used in implementing network intrusion in the cloud and explain in short MSP? ✓

FAR WESTERN UNIVERSITY
Final Examination-2077
Cloud Computing (CSIT 424.4)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: Eighth

Full Marks: 80

Time: 2hrs. 40 minutes

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group - B

Attempt any six questions.

6×8 = 48

1. ✓ Explain in short different participants of cloud.
2. ✓ What are benefits of using cloud models?
3. ✓ Explain advantages of using CaaS model.
4. Explain characteristics of Amazon EC2 service.
5. ✓ What are different building blocks of data centers?
6. ✓ Explain Service Oriented Architectures (SOA) with its characteristics.
7. Explain in short cloud security architecture design.
8. Explain different approaches for host security.

Group - C

Attempt any two questions.

2×16 = 32

9. ✓ What is cloud computing? Explain the Platform-as-a-service (PaaS) in detail.
10. What are different challenges in cloud security and explain cloud security control factors.
11. ✓ Explain Jericho cloud cube model.

FAR WESTERN UNIVERSITY
Semester End Examination-2080
Cloud Computing (CSIT.424.4)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: Eighth

Full Marks: 100

Time: 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group – 'A'

Attempt all questions (Very short answer questions)

8×3 = 24

1. What is cloud technology?
2. Mention some of the popularly used cloud computing services?
3. Write short note on host security.
4. Explain difference between traditional corporate data center and cloud data center.
5. What are the major difference between parallel and distributed computing?
6. Write short note on Monitoring-as-a-Service (MaaS).
7. What are different factors involved in disaster recovery planning.
8. What is distributed computing in grid.

Group – 'B'

Attempt any five questions. (Short answer questions)

5×8=40

9. What are the security issues which should be discussed with a cloud-computing vendor?
10. Explain disaster recovery process in cloud computing.
11. What are the key characteristics of PaaS?
12. Discuss the principal characteristics of SOA?
13. What are the SaaS Implementation Issues?
14. Explain the evolution of cloud computing.

P.T.O.

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Group – 'C'

Attempt any three questions. (Long answer questions)

3×12 = 36

- ✓ 15. What are the benefits of using cloud models? Explain the cloud computing deployment models (Types) and also discuss the ethical issues in cloud computing.
- ✓ 16. Describe the different building blocks required to build a data center?
17. What are different approaches used in cloud security control? Explain the purpose of a network intrusion detection system.
- ✓ 18. Write short notes on disaster management, benefits of data center virtualization, elastic IP addressing, IaaS.

FAR WESTERN UNIVERSITY
Semester End Examination-2079
Advanced Database Design (CSIT.423)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: Eighth

Shiv

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Group – A

Attempt all questions (Very Short Questions)

8×3=24

1. Define transaction.
2. List out any three differences between DBMS and RDBMS.
3. Define memory hierarchy?
4. Why do we use outer join?
5. Mention the role of indexing?
6. What are the characteristics of memory?
7. How do you recover data from loss?
8. Is DBMS useful for research?

Group – B

Attempt any five questions. (Short Questions)

5×8=40

9. What is two phases locking? Explain with example.
10. Mention the factors to evaluate indexing. Explain primary and ordered indexing with example.
11. What is query optimization? Write down its importance in brief.
12. Define ACID properties of a transaction with regard to DBMS.
13. Explain various types of failures in DBMS.
14. Define RAID. Explain this technology up to three levels.

Group – C

Attempt any three questions. (Long Questions)

3×12=36

15. What is file organization? Explain various file organizations with their merits and demerits.
16. Describe the query processing in detail.
17. What are the advantages of DBMS over file system? Explain in brief.
18. Write short notes on
 - a. Time stamp ordering protocol.
 - b. Deadlock prevention

Read (K) - (LW)
(LW)
(LR)

FAR-WESTERN UNIVERSITY
End-Term Examination-2074
Advanced Database Design (CSIT:423.2)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: Eighth

Shonkar Datt Bhatt

Full Marks: 80

Time: 2hrs. 40 minutes

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group - B

Attempt any six questions.

6×8 = 48

- ✓ 1. Explain four desirable properties of a transaction. Describe transaction state diagram in brief.
- ✓ 2. What is lock? Explain shared and exclusive locks with example.
- ✓ 3. Explain Sparse, Dense and Multi level index with example.
- ✓ 4. What is database recovery? Explain shadow paging with example.
5. Explain the importance of data integrity in DBMS.
- ✓ 6. What is a deadlock? How are deadlocks prevented?
- ✓ 7. List out factors for measuring performance of disk. Explain hashing in brief.
8. List out factors for measuring cost of query. Describe transformation rule with example.

Group - C

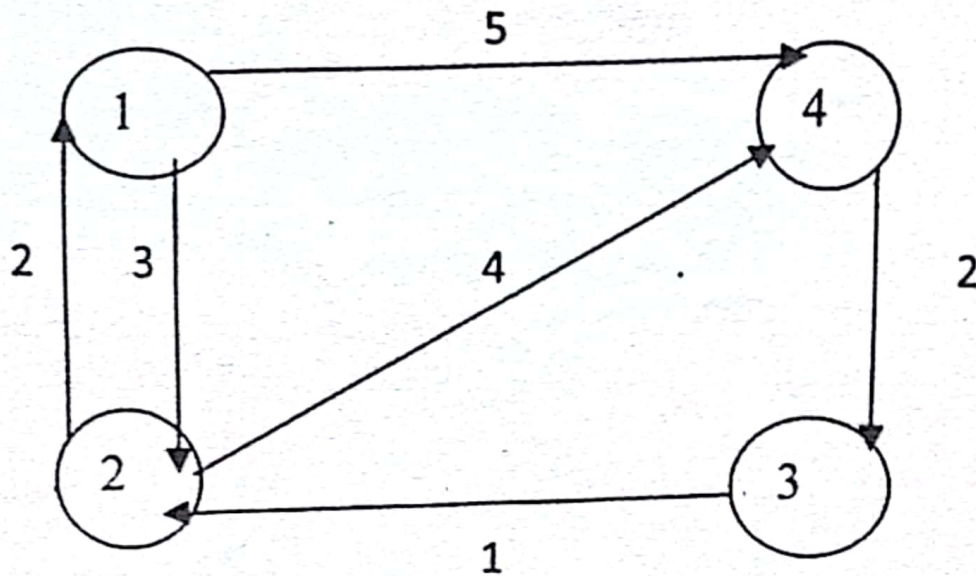
Attempt any two questions.

2×16 = 32

- ✓ 9. What is query processing? Explain steps in query processing with diagram and example.
- ✓ 10. Compare various file organizations with example. Also mention storage device hierarchy in brief.
11. Explain RAID technology up to three levels with example.

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10. What do you mean by combinatorial search? Explain alpha-beta pruning in detail.
11. Explain Floyd-Warshall algorithm. Use this algorithm to find the shortest path between all pairs of vertices of figure given below:



FAR-WESTERN UNIVERSITY
End-Term Examination-2076
Advanced Database Design (CSIT.423.2)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: Eighth

Full Marks: 80

Time: 2hrs. 40 minutes

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group - B

Attempt any six questions.

6×8 = 48

1. What are the advantages of DBMS over file system? Describe in brief.
2. Explain join dependency with example.
3. How can you convert an ER diagram into table? Discuss with example.
4. ✓ Explain Sparse, Dense and Multi level index with example.
5. ✓ Explain RAID technology up to three levels with example.
6. ✓ Explain state diagram of transaction process in brief.
7. ✓ Explain the data integrity in brief.
8. Write short notes (any two)
 - a. ✓ Data warehouse
 - b. 2PL
 - c. ✓ Shadow paging

Group - C

Attempt any two questions.

2×16 = 32

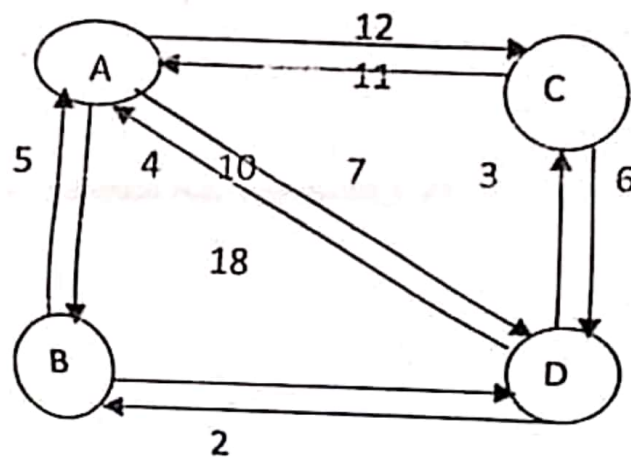
9. What is time stamping? How can it be used for concurrency control? Describe with the help of example.
10. ✓ What is Query processing? Describe query processing mechanism in details.
11. List out factors for measuring performance of disk. Compare various file organizations with example. Also mention storage device hierarchy in brief.

Group - C

Attempt any two questions.

2×16 = 32

9. Define clustering. What are the differences between parallel computing, distributed computing and cloud computing?
10. How can you differentiate linear equations with nonlinear equations? Use Naïve Gauss elimination to solve
$$20x_1 + 15x_2 + 10x_3 = 45$$
$$-3x_1 - 2.249x_2 + x_3 = 1.751$$
$$5x_1 + x_2 + 3x_3 = 9$$
11. Define combinatorial search. Solve the travelling salesman problem for the following graph by using the branch and bound algorithm.



FAR WESTERN UNIVERSITY

Final Examination-2077

Advanced Database Design (CSIT.423.2)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: Eighth

Full Marks: 80

Time: 2hrs. 40 minutes

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group - B

Attempt any six questions.

6×8 = 48

1. ✓ Explain join dependency with example?
2. ✓ Mention the desirable properties of transaction. Also describe state diagram of transaction.
3. ✓ Explain sparse and dense index with example.
4. ✓ What are the advantages of DBMS over file system?
5. ✓ What is deadlock? How are deadlocks prevented?
6. Differentiates between physical and logical data independence.
7. How do you increase the performance of database tuning?
8. ✓ Why do we need RAID? Explain RAID technology up to two levels with example.

Group - C

Attempt any two questions.

2×16 = 32

9. ✓ Describe the importance of data integrity in DBMS.
10. ✓ What is query processing? Explain steps in query processing with diagram and example.
11. Differentiate between sequential and random file organization. Explain storage hierarchy in brief.

FAR-WESTERN UNIVERSITY
End-Term Examination-2075
Advanced Database Design (CSIT.423.2)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: Eighth

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Full Marks: 80

Time: 2hrs. 40 minutes

Group - B

6×8 = 48

Attempt any six questions.

1. What is DBMS? List out the functions of DBMS in brief.
2. Explain RAID technology with example.
3. Describe the importance of concurrency control and explain any one method in brief.
4. Explain primary and secondary index with example.
5. What is deadlock? What are the necessary conditions for deadlock? Describe one method for deadlock recovery.
6. What is query optimization? Describe query optimization process in brief.
7. Formulate a concurrency control protocol based on timestamp-ordering protocol
8. Write short notes on.
 - a. Measures of Query Cost
 - b. Fixed length records and variable length records.

Group - C

2×16 = 32

Attempt any two questions.

9. What is query processing? Explain steps in query processing with diagram.
10. What is file organization? Explain various file organization with their advantages and disadvantages.
11. a. What are the various types of failures that occur in database? Discuss the log based recovery mechanism.
b. Describe the properties of transactions in brief.

9/11

FAR WESTERN UNIVERSITY
Semester End Examination-2080
Advanced Database Design (CSIT.423.2)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: Eighth

Full Marks: 100

Time: 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group – 'A'

Attempt all questions (Very short answer questions)

8×3 = 24

1. What is schema of a database?
2. Define foreign key.
3. What is hashing?
4. Write the statement of second normal form.
5. What are the performance measures of disks?
6. What are the steps in query decomposition?
7. What is multi-level index?
8. What is inheritance? How it is useful?

Group – 'B'

Attempt any five questions. (Short answer questions)

5×8=40

9. What is check point in database recovery? How does it help in database recovery?
10. Explain the life cycle of a transaction with state diagram.
11. What is indexing? What are the factors for evaluating indices? Explain primary indexing in brief.
12. Describe join dependency and fifth normal form (5NF) with example.
13. Explain deadlock and starvation. Discuss time stamp based protocol for concurrency control.
14. What are the advantages of using DBMS over file processing system?

");

Group – 'C'

Attempt any three questions. (Long answer questions)

3×12 = 36

15. What is query processing? Explain steps involved in query processing with example.
16. Explain RAID (Redundant Array of Independent Disks). Discuss pros and cons of RAID 0 and RAID 1 Level.
17. Discuss various types of file organization (any three) with their advantages in brief.
18. Write short notes on (any two)
 - A. Fixed length and variable length records
 - B. Memory Hierarchy
 - C. Data independence
